



L11 : Richardson Extrapolation

* To deal with the error of differentiation.
* only applicable for **central differentiation**

$$\text{Truncation error} = \left| \frac{f'''(\xi)h^2}{3!} \right| \quad O(h^2)$$ quadratic approach

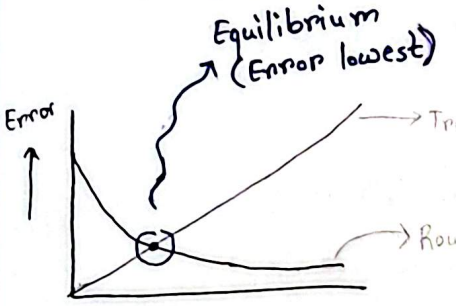
Now,

$$\varepsilon_m \left| \frac{f(x_0+h) - f(x_0-h)}{2h} \right|$$ → Rounding error

So;

→ Error $\propto h^2$ ← Truncation error

Error $\propto \frac{1}{h}$ ← Rounding error



$$f'(x) = D_h = \frac{f(x+h) - f(x-h)}{2h}$$

Exam:

$h = \frac{h}{2}$ এবার value দিবে

$$D_{h/2} = \frac{f(x+\frac{h}{2}) - f(x-\frac{h}{2})}{2\frac{h}{2}}$$

Now,

$$f(x+h) - f(x-h)$$

$$= 2f'(x)h + 2\frac{f^3(x)h^3}{3!} + 2\Theta(h^5)$$

Taylor series:

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2} + \frac{f'''(x)h^3}{3!} + \Theta(h^4)$$

Now,

$$D_h = \frac{f(x+h) - f(x-h)}{2h}$$

$$= \frac{1}{2h}\left(2f'(x)h + \frac{2f^3(x)h^3}{3!} + 2\Theta(h^5)\right)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)h^2}{2!} - \frac{f'''(x)h^3}{3!} + O(h^4)$$

$$= f'(x) + \frac{f^3(x)h^2}{3!} + \Theta(h^4)$$

Derivation (Derive term) — Truncation error (Error term)

$$D_h = f'(x) + \frac{f^3(x)h^2}{3!} + \frac{f^5(x)h^4}{5!} + \Theta(h^6)$$

Derivation term

কমাতে error
বেশি হয়ে যায়

↓ error $\propto \downarrow h^2, h^4$

- Prove Richardshon extrapolation with $\frac{2h}{2}$
- " " " $\frac{h}{2}$

$$D_{h/2} = f'(x) + \frac{f^3(x)}{3!}\frac{h^2}{4} + \frac{f^5(x)}{5!}\frac{h^4}{16} + \theta(h^6)$$

- Using $h/2$ get rid of the term that causes a lot of error

~~$4D_{h/2} - D_h = 4f'(x) + \frac{f^3(x)h^2}{3!} + \frac{f^5(x)h^4}{5!}$~~

$$4D_{h/2} - D_h = 4f'(x) + \frac{f^3(x)h^2}{3!} + \frac{4f^5(x)h^4}{5!\times 16} + 4\theta(h^6)$$

$$- f'(x) + \frac{f^3(x)h^2}{3!} + \frac{f^5(x)h^4}{5!} + \theta(h^6)$$

$$= 3f'(x) + \left(\frac{1}{4} - 1\right)\frac{f^5(x)h^4}{5!} + \theta(h^6)$$

$$\Rightarrow \frac{4D_{h/2} - D_h}{3} = f'(x) - \frac{1}{4}\frac{f^5(x)}{5!}h^4 + \theta(h^6)$$ [1st order differentiation]

↓
Derive term এর coefficient থাকবে না

2nd order ⟶ $O(h^6)$ পর্যন্ত যাব

3rd order ⟶ $O(h^8)$ পর্যন্ত যাব

Lets do $D_h^{(2)}$:

$$D_h^{(1)} = f'(x) - \frac{1}{4}\frac{f^5(x)\,h^4}{5!} + O(h^6)$$

$$D_{h/2}^{(1)} = f'(x) - \frac{1}{4}\frac{f^5(x)}{5!}\frac{h^4}{16} + O(h^6)$$

$$16\,D_{h/2} - D_h^{(1)} = 15f'(x) + O(h^6)$$

$$\frac{16\,D_{h/2} - D_h^{(1)}}{15} = f'(x) + O(h^6) = D_h^{(2)}$$

↳ Richardson Extrapolation's second order differentiation

$$\frac{2^p\,D_{h/2}^{t+1} - D_{h/2}^{t-1}}{2^p - 1}$$ [p ⟶ যে h remove করবো]

যদি বলে $D_h^{(1)}$ using $h = \frac{3h}{2}$?

Ans: $\frac{4}{9} D_{3h/2} - D_h$ [Try]

$$D_h = f'(x) + f^3(x)\frac{h^2}{3!} + O(h^5)$$

$$D_{3h/2} = f'(x) + f^3(x)\frac{h^2}{3!} \times \frac{9}{4} + O(h^5)$$

$$\therefore \frac{4}{9} D_{\frac{3h}{2}} - D_h$$

$f(x) = xe^{-3x} + x^2$ (a) Compute $D_{0.1}^{(1)}$ at $x_0 = 2$

(b) Find relative error

(a)

$$\text{Richardson} = \frac{4D_{h/2}^{(0)} - D_h^{(0)}}{3}$$

$$= \frac{4D_{0.05}^{(0)} - D_{0.1}^{(0)}}{3}$$

$$D_{0.05}^{(0)} = \frac{f(x_0+h) - f(x_0-h)}{2h}$$

2.05
1.95

$$= \frac{4.20 - 3.808}{2 \times 0.05} = 3.92$$

$$D_{0.1}^{(0)} = \frac{4.41 - 3.616}{2 \times 0.1} = 3.97$$

Now, $\frac{4 \times 3.92 - 3.97}{3} = 3.90$

(b)

Actual value: $f(x) = xe^{-3x} + x^2$

$$f'(x) = e^{-3x} - 3xe^{-3x} + 2x$$

$$= (1-3x)e^{-3x} + 2x$$

$$f'(2) = 3.98$$

$$\therefore |3.90 - 3.98|$$